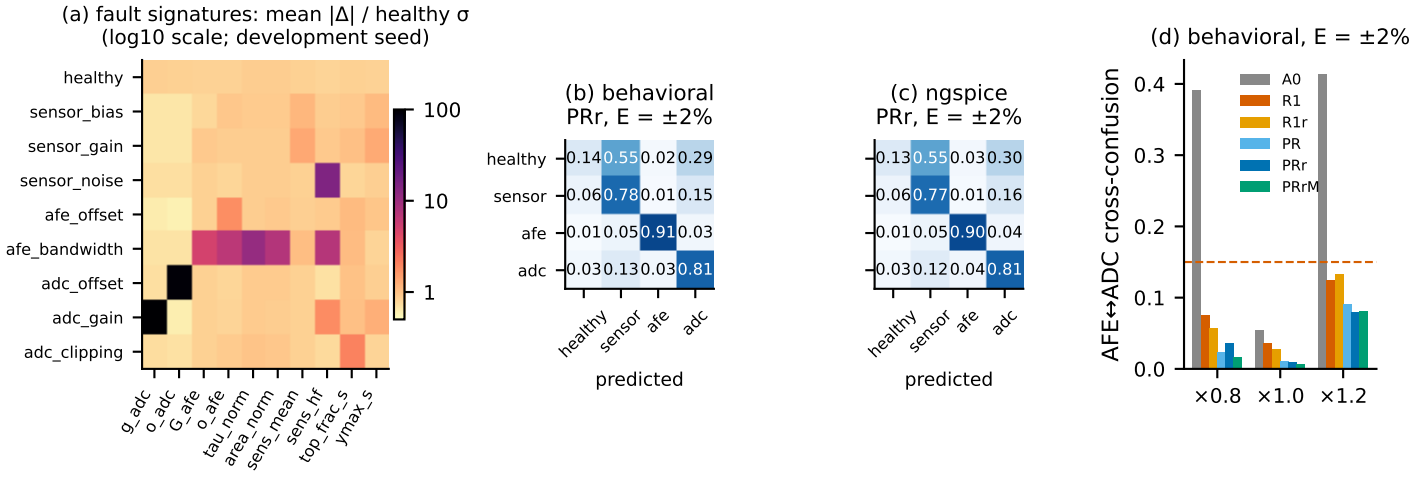
(a) fault signatures: mean |Δ| / healthy σ
(log10 scale; development seed)
healthy
sensor_bias
sensor_gain
sensor_noise
afe_offset
afe_bandwidth
adc_offset
adc_gain
adc_clipping
100
10
1
g_adc
o_adc
G_afe
o_afe
tau_norm
area_norm
sens_mean
sens_hf
top_frac_s
ymax_s
(b) behavioral
PRr, E = ±2%
healthy 0.14 0.55 0.02 0.29
sensor 0.06 0.78 0.01 0.15
afe 0.01 0.05 0.91 0.03
adc 0.03 0.13 0.03 0.81
healthy
sensor
afe
adc
predicted
(c) ngspice
PRr, E = ±2%
healthy 0.13 0.55 0.03 0.30
sensor 0.06 0.77 0.01 0.16
afe 0.01 0.05 0.90 0.04
adc 0.03 0.12 0.04 0.81
healthy
sensor
afe
adc
predicted
(d) behavioral, E = ±2%
AFE↔ADC cross-confusion
0.0
0.1
0.2
0.3
0.4
A0
R1
R1r
PR
PRr
PRrM
×0.8
×1.0
×1.2